NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-94/6C
19 June 1985
SUPERSEDING
MIL-R-94/6B
27 August 1974

PERFORMANCE SPECIFICATION

RESISTORS, VARIABLE, COMPOSITION, STYLE 2RV7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style 2RV7 composition variable, dual (2 ganged) resistors.

1.2 Classification.

1.2 Type designation. The type designation shall be in the following form:

2RV7	N	Y	SD	103	102	A
Style (1.2.1.1)	Bushing (1.2.1.2)	Temperature and moisture resistance characteristic (1.2.1.3)	Operating shaft (1.2.1.4)	Resistance (panel) section)	Resistance (rear section) (1.2.1.5)	Resistance characteristic (1.2.1.6)

The type designation shall be listed in the order in which the sections (cups) are stacked (ganged) beginning with the section (cup) nearest the mounting surface.

1.2.1.1 Style. The style is identified by the two letter symbol "RV" preceded and followed by a one-digit number. The first number identifies a dual ganged (2 cup) resistor, the letters identify composition, variable resistors and the last number identifies the size and power rating of both cups (see 3.6).

1.2.1.2 Bushing. The type of bushing is identified by a single letter in accordance MIL-R-94.

1.2.1.3 Temperature and moisture resistance characteristic. The temperature and moisture resistance characteristic is identified by a single letter in accordance with MIL-R-94.

1.2.1.4 Operating shaft. The operating shaft styles and lengths are identified by a two-letter symbol. The first letter indicates operating shaft style in accordance with MIL-R-94, and the second letter indicates operating shaft length as specified in table I.

1.2.1.5 Resistance. The nominal total resistance value expressed in ohms is identified by a three-digit number. The first two digit represent significant figures and the last specifies the number of zeroes to follow. The first three digits in the type designation is the value of resistor cup nearest the mounting surface and the next three digits is the resistance value of the second cup.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: DESC-ELDM, 1507 Wilmington Pike, Dayton, OH 45444-5765 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5905

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TABLE I. Length of operating shaft.

Symbol	Length of shaft from mounting surface of resistors. Nominal 1/	
	Bushings M and S	Bushings L and T
	inches ±.031 (.79 mm)	inches ±.031 (.79 mm)
A- - - -		.625 (15.88 mm)
B- - - -	.500 (12.70 mm)	
D- - - -	.875 (22.23 mm)	.875 (22.23 mm)
G- - - -	1.250 (31.75 mm)	
M- - - -	1.500 (38.10 mm)	
J- - - -	2.000 (50.80 mm)	
K- - - -	2.500 (63.50 mm)	

1/ For resistors with panel seals, the mounting surface shall be considered the metal face of the panel seal when the seal is firmly seated against the resistor body.

1.2.1.6 Resistance characteristic combination. The resistance characteristic combination is identified by a single letter which describes a combination of resistance characteristic symbols per MIL-R-94, as specified in table II.

TABLE II. Resistance characteristic combination symbol.

Resistance characteristic combination symbol	Panel section Resistance characteristic symbol	Rear section Resistance characteristic symbol
A	A	A
C	C	C
G	A	C
M	C	A

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-94 - Resistors, Variable, Composition, General Specification for.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-94, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Mounting and locking nuts. The mounting nut shall be .094 inch (2.39 mm) thick and, unless otherwise specified, shall measure .500 inch (12.70 mm) across the hexagonal flats. The locking nuts shall be within the limits of .151 inch (3.84 mm)

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and .234 inch (5.94 mm) thick and shall measure .500 inch (12.70 mm) across the hexagonal flats. Thread size shall be .375-32NEF-2B.

3.2.2 Internal-tooth lockwasher. An internal-tooth lockwasher of suitable size shall be supplied.

3.2.3 Retainer rings. If retainer rings are used they shall be not thicker than 0.047 inch (1.19 mm).

3.3 Total mechanical rotation. The total rotation shall be within the limits of 309° and 320°.

3.4 Shaft length. The length of operating shaft shall be in accordance with table I.

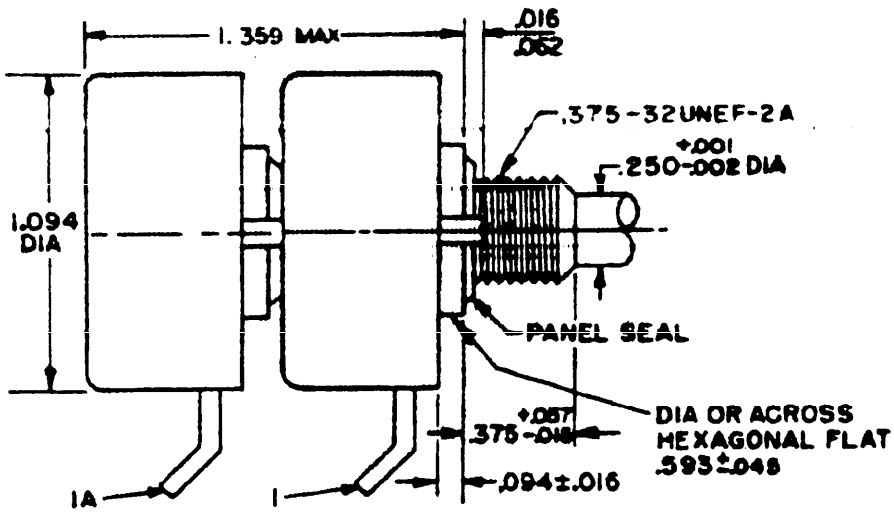
3.5 Nominal total resistance and rated continuous working voltage. The nominal total resistance and rated continuous working voltage (RCWV) shall be as specified in table III. These voltage ratings are based on full power dissipation in the specified section. For other combinations of power ratings listed in 3.6, the applicable working voltages should be calculated from the formula $E = \sqrt{PR}$.

TABLE III. Nominal total resistance and rated continuous working voltage.

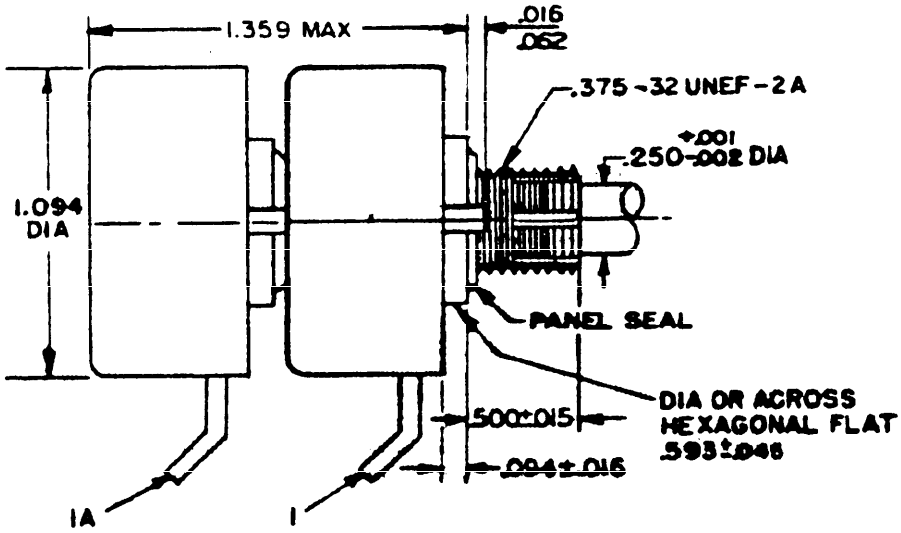
Nominal total resistance	RCWV 1/ (volts)							
	Resistance characteristic combination							
	A		C		G		H	
ohms	Panel section	Rear section	Panel section	Rear section	Panel section	Rear section	Panel section	Rear section
50	10	9	---	---	10	---	---	9
100	14	13	10	9	14	9	10	13
150	17	15	12	11	17	11	12	15
200	20	18	---	---	20	---	---	18
250	22	20	16	14	22	14	16	20
350	26	24	18	17	26	17	18	24
500	32	28	23	20	32	20	23	28
750	39	35	28	24	39	24	28	35
1,000	45	40	32	28	45	28	32	40
1,500	55	49	39	35	55	35	39	49
2,000	63	57	45	40	63.3	40	45	57
2,500	71	63	50	45	71	45	50	63
3,500	84	75	60	53	84	53	60	75
5,000	100	89	71	63	100	63	71	89
7,500	122	110	87	77	122	77	87	110
10,000	141	126	100	89	141	89	100	126
15,000	173	155	123	109	173	109	123	155
20,000	200	179	141	126	200	126	141	179
25,000	224	200	160	141	224	141	160	200
35,000	264	237	187	167	264	167	187	237
50,000	316	283	224	200	316	200	224	283
75,000	387	346	249	245	387	245	249	346
.10M	445	400	316	283	445	283	316	400
.15M	500	490	350	346	500	346	350	490
.20M	500	500	350	350	500	350	350	500
.25M	500	500	350	350	500	350	350	500
.35M	500	500	350	350	500	350	350	500
.50M	500	500	350	350	500	350	350	500
.75M	500	500	350	350	500	350	350	500
1.0M	500	500	350	350	500	350	350	500
1.5M	500	500	350	350	500	350	350	500
2.0M	500	500	350	350	500	350	350	500
2.5M	500	500	350	350	500	350	350	500
3.5M	500	500	350	350	500	350	350	500
5.0M	500	500	350	350	500	350	350	500

1/ Rated continuous working voltage at 70°C.
These are maximum values that would apply only when the other section has zero wattage dissipated.

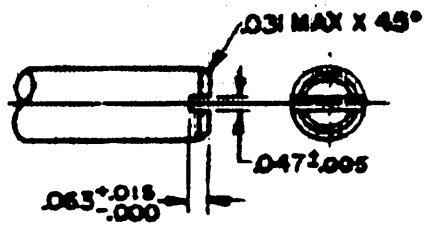
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SHAFT-AND PANEL- SEALED - STANDARD BUSHING TYPE



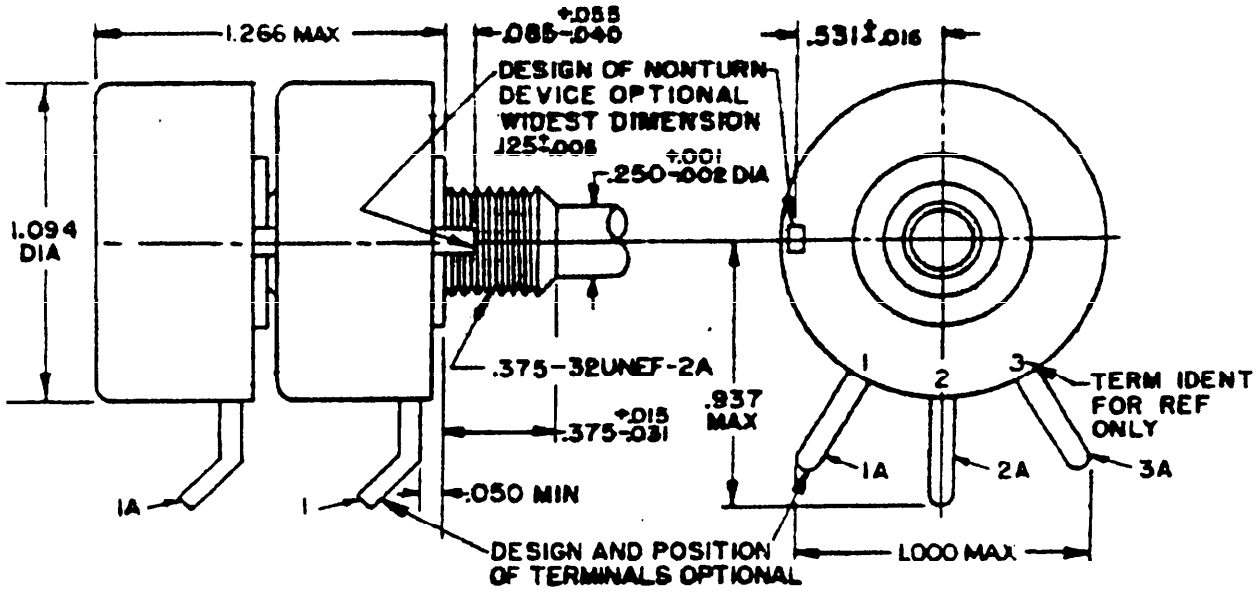
SHAFT-AND PANEL-SEALED-LOCKING BUSHING TYPE



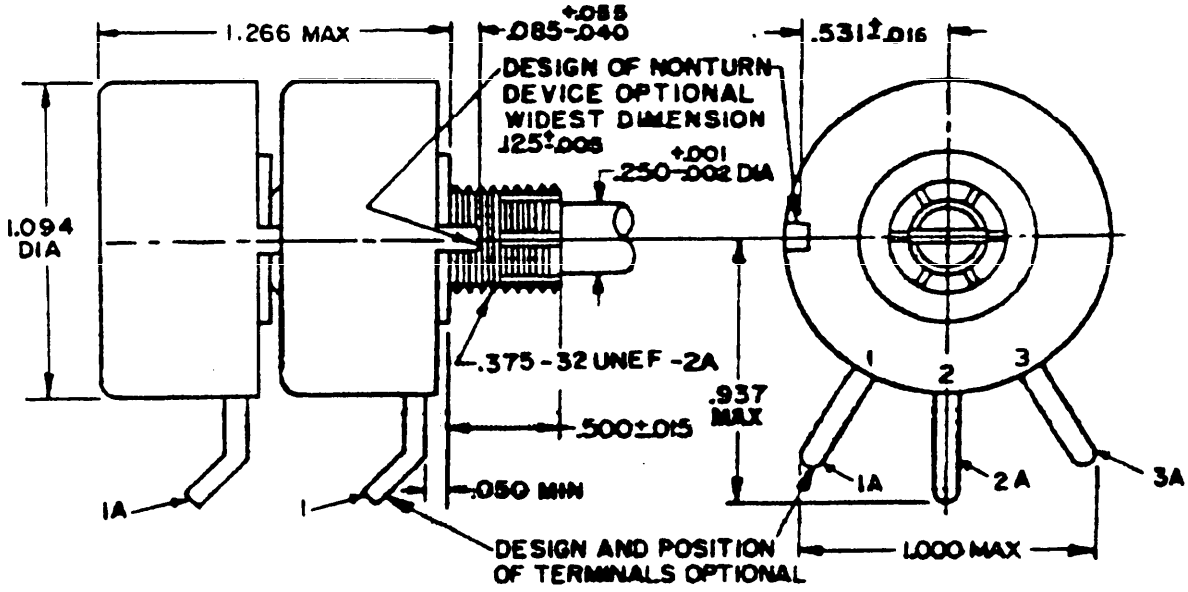
SLOTTED SHAFT

FIGURE 1. Style 2RV7 resistors.

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STANDARD-BUSHING TYPE



LOCKING-BUSHING TYPE

FIGURE 1. Style 2RV7 resistors - Continued.

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INCHES	MM	INCHES	MM
.001	0.03	.063	1.60
.002	0.05	.085	2.16
.005	0.13	.094	2.39
.015	0.38	.125	3.18
.016	0.41	.250	6.35
.031	0.79	.375	9.53
.040	1.02	.500	12.70
.045	1.14	.531	13.49
.047	1.19	.593	15.06
.050	1.27	.937	23.80
.055	1.40	1.000	25.40
.057	1.45	1.094	27.79
.062	1.57	1.266	32.16
		1.359	34.52

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.062 (1.57 mm).
4. The pictorial view of the styles are given as representative of the envelope of the items. Slight deviations from the outline shown, which are contained within the envelope, and do not alter the functional aspects of the device are acceptable.

FIGURE 1. Style 2RV7 resistors - Continued.

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3.6 Power rating. The maximum continuous power rating in watts with both entire resistor elements in the circuit for the various resistance characteristic combinations are as specified in table IV.

TABLE IV. Maximum continuous power rating (watts).

Resistance characteristic combination							
A		C		G		H	
Panel resistor	Rear resistor	Panel resistor	Rear resistor	Panel resistor	Rear resistor	Panel resistor	Rear resistor
0	1.6	0	.8	0	.8	0	1.6
.2	1.59	.1	.79	.2	.79	.1	1.59
.4	1.56	.2	.78	.4	.78	.2	1.56
.6	1.52	.3	.76	.6	.76	.3	1.52
.8	1.46	.4	.73	.8	.73	.4	1.46
1.0	1.38	.5	.69	1.0	.69	.5	1.38
1.2	1.28	.6	.64	1.2	.64	.6	1.28
1.4	1.14	.7	.57	1.4	.57	.7	1.14
1.6	.96	.8	.48	1.6	.48	.8	.96
1.8	.70	.9	.35	1.8	.35	.9	.70
2.0	0	1.0	0	2.0	0	1.0	0

3.7 Torque.

3.7.1 Operating. The torque required to effect rotation shall be not less than 1 ounce-inch nor more than 12 ounce-inches.

3.7.2 Stop. The torque applied through the operating shaft to the stops shall be 8 pound-inches.

3.7.3 Locking bushing. For the locking-bushing type, the locking nut shall be tightened with a torque of 10 pound-inches. After the locking nut is tightened, the contact arm shall not move when a torque of 30 ounce-inches is applied to the shaft.

3.8 Thermal cycling. The resistors shall not change in resistance in excess of 4 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-94, and as specified herein.

4.2 Dielectric withstanding voltage. In the dielectric withstanding voltage test, the applied potential at reduced barometric pressure shall be 450 volts, rms.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-94.

6. NOTES

6.1 Notes. The notes specified in MIL-R-94 are applicable to this specification.

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - ER, MI, MU
Navy - AS, OS
Air Force - 17, 85, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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